📘 IBM HR Analytics – Employee Attrition Prediction Using Machine Learning

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ABSTRACT

Employee attrition is a major concern for organizations as it affects productivity, cost, and workforce stability. This project aims to analyze employee data and build a predictive model to determine whether an employee is likely to leave the company.

Using the IBM HR Analytics dataset, data preprocessing, visualization, and Logistic Regression modeling were performed. The model was evaluated using accuracy, confusion matrix, and classification report.

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1. INTRODUCTION

Employee attrition refers to employees leaving an organization voluntarily or involuntarily. High attrition rates can increase recruitment costs and reduce organizational efficiency.

By applying Machine Learning techniques, companies can predict employee attrition and take preventive actions to improve retention.

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2. DATASET DESCRIPTION

The dataset contains employee-related information such as:

- Age
- Job Role
- Monthly Income
- Years at Company
- Job Satisfaction
- Overtime
- Attrition (Target Variable)

The target variable **Attrition** contains:

- Yes (Employee Left)
- No (Employee Stayed)

It was converted into:

- 1 → Yes
- 0 → No

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3. OBJECTIVES

- To preprocess and clean HR dataset
- To perform exploratory data analysis
- To encode categorical variables
- To build a Logistic Regression classification model
- To evaluate model performance

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4. METHODOLOGY

4.1 Data Cleaning

Unnecessary columns were removed:

```
df.drop(['EmployeeCount','EmployeeNumber','Over18','StandardHours'], axis=1, inplace=True)
```

Target column was converted:

```
df['Attrition'] = df['Attrition'].map({'Yes':1, 'No':0})
```

4.2 Encoding Categorical Variables

Categorical columns were encoded using Label Encoder:

```
le = LabelEncoder()
```

This converts text categories into numerical form for model training.

4.3 Feature Scaling

StandardScaler was applied to normalize data before training Logistic Regression.

```
scaler = StandardScaler()
```

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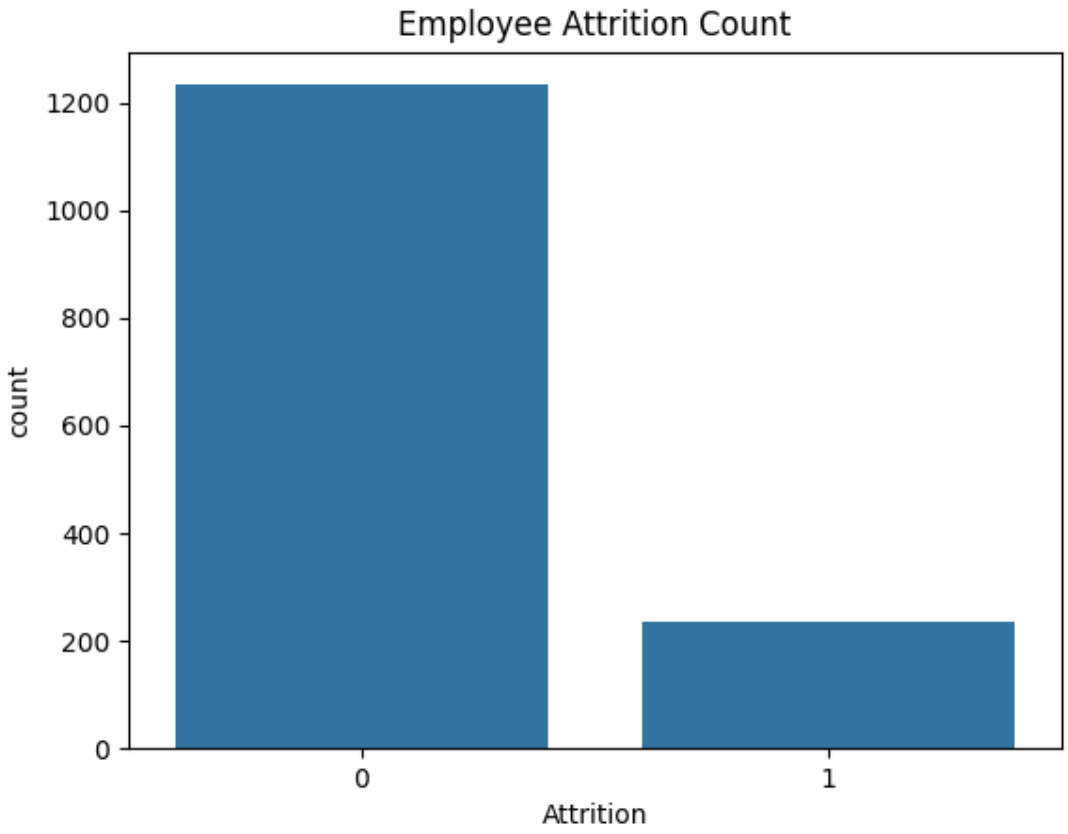
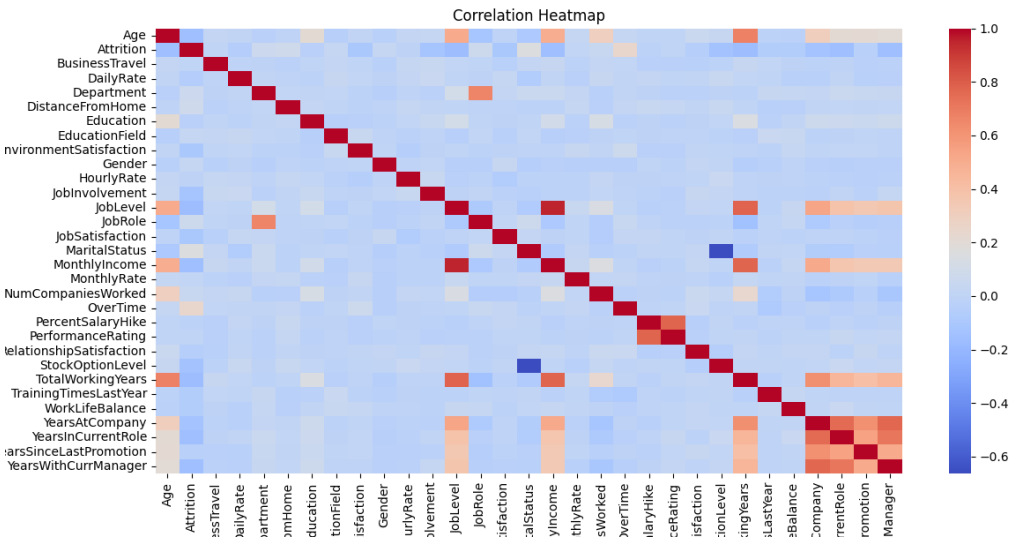
5. EXPLORATORY DATA ANALYSIS (EDA)

📊 **Figure 1: Employee Attrition Count**



This graph shows the number of employees who left versus those who stayed.
It helps understand class imbalance in the dataset.

📊 Figure 2: Correlation Heatmap



The heatmap shows correlation between different variables.
It helps identify which features strongly influence attrition.

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6. MODEL BUILDING

The dataset was divided into training and testing sets (80%-20%).

Logistic Regression model was used:

model = LogisticRegression(max_iter=5000, class_weight='balanced')

Class weight was set to balanced to handle class imbalance.

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7. MODEL EVALUATION

Predictions were generated:

y_pred = model.predict(X_test)

7.1 Accuracy

Accuracy measures the percentage of correctly predicted instances.

Paste your output value here:

Accuracy = 0.7142857142857143 %

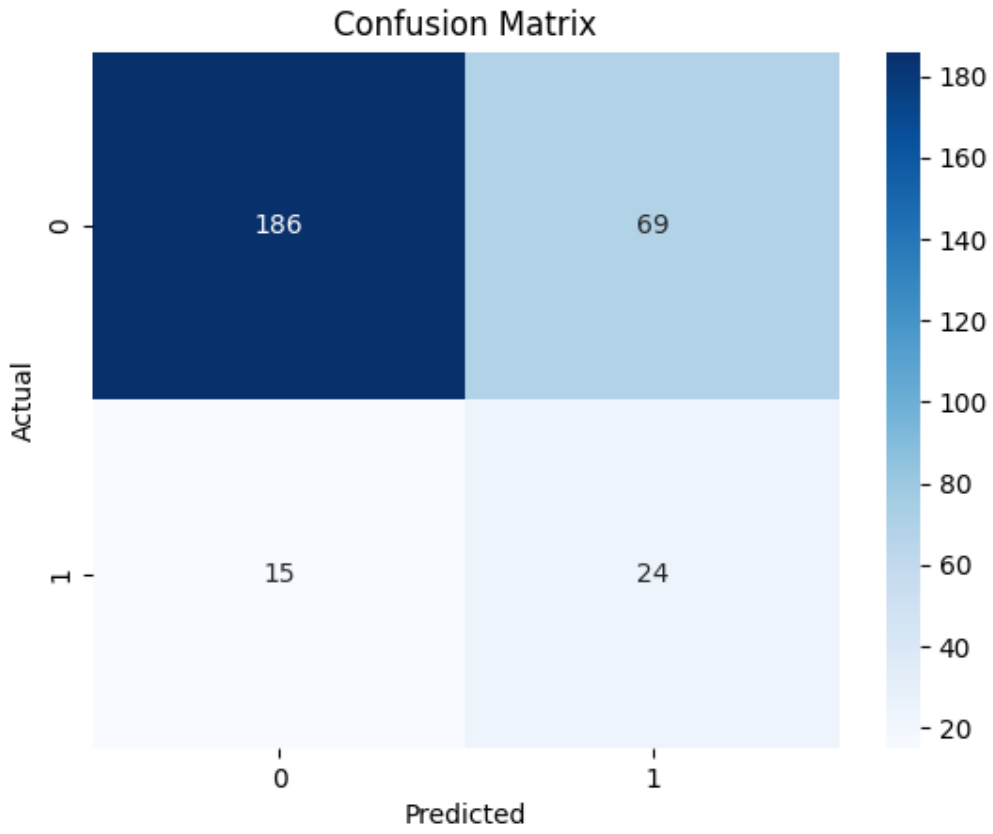
7.2 Classification Report

Includes:

- Precision
- Recall
- F1-Score
- Support

These metrics help evaluate classification performance.

📊 **Figure 3: Confusion Matrix**



The confusion matrix shows:

- True Positives
- True Negatives
- False Positives
- False Negatives

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8. RESULTS AND DISCUSSION

- The model successfully classifies employees likely to leave.
- Logistic Regression performs well for binary classification problems.
- Important features influencing attrition include overtime, income, and years at company.

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9. CONCLUSION

This project demonstrates how Machine Learning can be used to predict employee attrition.

By identifying employees at risk of leaving, organizations can take preventive actions such as improving work conditions, increasing employee engagement, and revising compensation strategies.

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10. FUTURE SCOPE

- Use advanced models like Random Forest or XGBoost
- Perform hyperparameter tuning
- Apply SMOTE for better class balancing
- Deploy model as HR decision-support system

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11. TOOLS AND TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn